

■ POWER BI Interview Questions & Answers

For Freshers — Simple English, Easy to Remember

50+ Most-Asked Power BI Questions with Clear Answers

SECTION 1 — Power BI Basics

Q1. What is Power BI?

Answer: Power BI is Microsoft's tool for creating interactive charts, reports, and dashboards from data. It helps businesses visualise their data and make data-driven decisions. You connect it to data sources (Excel, SQL, etc.), clean the data, create relationships, and build visual reports.

Q2. What are the main parts (components) of Power BI?

Answer: Power BI Desktop — the app on your computer where you BUILD reports. Power BI Service — the website (app.powerbi.com) where you SHARE and VIEW reports online. Power BI Mobile — view reports on phone/tablet. Power BI Gateway — connects cloud Power BI to your company's local data.

Remember: Desktop = build. Service = share. Mobile = view on phone.

Q3. What is the difference between a Report and a Dashboard in Power BI?

Answer: Report = multiple pages of charts and visuals, built in Desktop, focuses on one dataset. Dashboard = a single page overview in Power BI Service, you PIN charts from reports to it, can have charts from multiple reports.

Remember: Report = detailed analysis. Dashboard = quick overview.

Q4. What are the data connectivity modes in Power BI?

Answer: Import Mode — data is COPIED into Power BI (like a snapshot). Fast performance. Data can be up to 1 GB. Must refresh to get new data. DirectQuery Mode — Power BI connects LIVE to the database. Every click sends a query. Always fresh data but slower. Composite — mix of both Import and DirectQuery.

Remember: For interview: Import = fast but not live. DirectQuery = live but slower.

Q5. What types of data sources can Power BI connect to?

Answer: Excel files, CSV files, SQL Server, MySQL, Oracle, Azure, Google Analytics, SharePoint, Web pages (URLs), JSON/XML files, Python/R scripts, Salesforce, and many more. In Desktop, click 'Get Data' to see all options.

Q6. What is Power Query in Power BI?

Answer: Power Query is Power BI's data cleaning tool. When you load data, Power Query lets you clean, reshape, filter, rename, merge, and transform data BEFORE it goes into your report. Each step is recorded and can be refreshed automatically.

Example: Remove blank rows, split columns, change data types — all done in Power Query.

Remember: Power Query = ETL tool inside Power BI (Extract, Transform, Load)

Q7. What is a calculated column vs a measure in Power BI?

Answer: Calculated Column: creates a new column in your table, calculated row by row, stored in the model, uses memory/storage. Example: Full Name = FirstName & ' ' & LastName. Measure: a calculation that runs when you drag it into a visual, not stored, calculated on the fly based on the visual's filter context. Example: Total Sales = SUM(Sales[Amount]).

Remember: Calculated Column = permanent new column. Measure = dynamic calculation.

Q8. What is DAX?

Answer: DAX stands for Data Analysis Expressions. It is the formula language used in Power BI to create calculated columns and measures. DAX looks similar to Excel formulas but is more powerful — it understands table relationships and filter context.

Example: Total Sales = SUM(Sales[Revenue]) — a simple DAX measure.

SECTION 2 — DAX Formulas

Q9. What is SUM, AVERAGE, COUNT, MIN, MAX in DAX?

Answer: SUM(table[column]) — adds all values. AVERAGE(table[column]) — calculates average. COUNT(table[column]) — counts non-empty values. COUNTA counts all values including text. COUNTROWS(table) counts all rows. MIN/MAX returns smallest/biggest value.

Example: Total Revenue = SUM(Sales[Revenue]) / Avg Order = AVERAGE(Orders[Amount])

Q10. What is CALCULATE in DAX?

Answer: CALCULATE is the most powerful DAX function. It evaluates an expression but lets you CHANGE the filters. Think of it as 'calculate this, but only for these specific conditions'.

Example: Sales in 2023 = CALCULATE(SUM(Sales[Revenue]), Sales[Year] = 2023) — sum of revenue only for 2023.

Remember: CALCULATE = change filter + calculate

Q11. What is the difference between FILTER and CALCULATE?

Answer: CALCULATE changes the filter context for the whole expression. FILTER creates a filtered TABLE (you then use it inside other functions). FILTER is usually used inside CALCULATE.

Example: CALCULATE(SUM(Sales[Amt]), FILTER(Sales, Sales[Region]='North')) — sum only for North.

Q12. What is SUMX? How is it different from SUM?

Answer: SUM adds values in one column. SUMX goes row by row, calculates something for each row, then adds up all the results. Use SUMX when you need to multiply or calculate something per row first.

*Example: Total Revenue = SUMX(Sales, Sales[Qty] * Sales[Price]) — multiplies qty*price for each row, then sums all.*

Remember: SUM = simple column sum. SUMX = row-by-row calculation then sum.

Q13. What is COUNTROWS?

Answer: COUNTROWS(table) counts the number of rows in a table. Very useful for counting orders, customers, etc.

Example: Total Orders = COUNTROWS(Orders) / Orders in 2023 = COUNTROWS(FILTER(Orders, Orders[Year]=2023))

Q14. What is RELATED in DAX?

Answer: RELATED() fetches a value from a related (connected) table. Used in calculated columns to bring in data from another table using the relationship.

Example: In Orders table: Category = RELATED(Products[Category]) — gets category from Products table.

Q15. What is IF in DAX?

Answer: IF(condition, value_if_true, value_if_false) — works like IF in Excel. Used to create conditional measures or columns.

Example: Status = IF(Sales[Amount] > 1000, 'High', 'Low') / Grade = IF(Marks[Score]>=90,'A', IF(Marks[Score]>=80,'B','C'))

Q16. What is DIVIDE? Why use it instead of / ?

Answer: DIVIDE(numerator, denominator, alternate_result) safely divides and returns the alternate_result (default = BLANK) if denominator is 0, instead of giving an error.

Example: Conversion Rate = DIVIDE(Converted, TotalVisitors, 0) — returns 0 if no visitors instead of error.

Remember: Always use DIVIDE for division in DAX — never just /

Q17. What is ALL() in DAX?

Answer: ALL() removes filters from a table or column. Used to calculate totals or percentages that ignore current filters.

Example: % of Total = DIVIDE(SUM(Sales[Rev]), CALCULATE(SUM(Sales[Rev]), ALL(Sales))) — gives each row as % of grand total.

Q18. What is DISTINCTCOUNT?

Answer: DISTINCTCOUNT(column) counts the number of unique values in a column. Different from COUNT which counts all non-empty values.

Example: Unique Customers = DISTINCTCOUNT(Orders[CustomerID]) — counts each customer only once.

Q19. What are Date/Time functions in DAX?

Answer: TODAY() — today's date. NOW() — current date and time. YEAR(date), MONTH(date), DAY(date) — extract parts. DATEDIFF(start, end, MONTH) — difference between dates. DATEADD(date, -1, MONTH) — go back 1 month. TOTALYTD — year to date total.

Example: Days Since Order = DATEDIFF(Orders[OrderDate], TODAY(), DAY)

Q20. What is Year-To-Date (YTD) measure?

Answer: YTD means from the start of the year to today. In DAX: YTD Sales = TOTALYTD(SUM(Sales[Revenue]), Dates[Date]). This requires a proper Date/Calendar table in your model.

Remember: TOTALYTD, TOTALQTD (quarter), TOTALMTD (month) are time intelligence functions.

SECTION 3 — Data Modelling

Q21. What is a Data Model in Power BI?

Answer: A data model is how your tables are connected (related) to each other. When you have multiple tables (Sales, Products, Customers), you connect them using relationships on common columns. This lets you use data from multiple tables in one chart.

Q22. What is a Star Schema?

Answer: Star Schema = one large central FACT table surrounded by smaller DIMENSION tables. Fact table = actual transactions/events (Sales, Orders). Dimension tables = descriptive information (Customers, Products, Dates). Called 'star' because it looks like a star shape.

Example: Fact_Sales table connected to Dim_Date, Dim_Product, Dim_Customer.

Remember: Star Schema is the best practice for Power BI models.

Q23. What is the difference between a Fact table and a Dimension table?

Answer: Fact table — contains numbers/measures (sales amount, quantity, revenue). Has foreign keys pointing to dimension tables. Usually has many rows. Dimension table — contains descriptive text information (customer name, product category, city). Has a primary key. Usually smaller.

Q24. What is a relationship in Power BI? What types exist?

Answer: Relationship = link between two tables on a common column. Types: One-to-Many (1:*) — most common, one row in dim matches many rows in fact. Many-to-Many (*:*) — use carefully, can cause issues. Active vs Inactive — only one active relationship can exist between two tables at a time.

Remember: Most relationships in a good model should be One-to-Many.

Q25. What is filter direction (single vs bidirectional)?

Answer: Single direction — filter flows one way (from dimension to fact). Recommended. Bidirectional — filter flows both ways. Can cause problems (circular filters). Use bidirectional only when absolutely needed.

Remember: Default = single direction. Bidirectional = use carefully.

Q26. What is a Date/Calendar table? Why is it needed?

Answer: A Date table is a dedicated table with one row for every date. Columns: Date, Day, Month, Month Name, Quarter, Year, Weekday etc. It's needed for time intelligence functions like YTD, MTD, QTD to work correctly. Best practice: always have a Date table.

SECTION 4 — Visuals & Reports

Q27. What types of visuals are in Power BI?

Answer: Bar/Column chart — compare categories. Line chart — trends over time. Pie/Donut — part of a whole. Map — geographical data. Card — single number (Total Sales: 1.2M). Table/Matrix — detailed data. Scatter — relationship between two numbers. Waterfall — step by step addition/subtraction. KPI visual — actual vs target.

Q28. What is a Slicer?

Answer: A slicer is an interactive filter button on your report. Users can click it to filter all visuals on the page. Example: a Year slicer lets users click 2022/2023/2024 and all charts update.

Remember: Slicer = interactive visual filter for the user.

Q29. What is the difference between a filter pane and a slicer?

Answer: Slicer — visible on the report canvas, users can see and click it. Filter Pane — hidden panel on the side, used by the report designer to set filters. Slicer is more user-friendly.

Q30. What is a Tooltip in Power BI?

Answer: A tooltip is a small pop-up that appears when you hover over a data point, showing extra information. You can create custom tooltip pages with multiple visuals that appear on hover.

Q31. What is Drill Down/Drill Through in Power BI?

Answer: Drill Down — go from higher level to lower level within the same visual. Example: Year → Quarter → Month → Day. Drill Through — right-click on a data point and jump to a detailed page about that item. Example: right-click on 'North Region' to see all North Region details.

Q32. What is Row Level Security (RLS) in Power BI?

Answer: RLS restricts data access for different users. Example: a regional manager should only see their region's data. You create roles with DAX filters (like [Region] = 'North'), assign users to roles in Power BI Service, and users can only see their allowed data.

Remember: RLS = data security per user/group.

Q33. How do you optimise a slow Power BI report?

Answer: 1) Use Import mode instead of DirectQuery if possible. 2) Remove unused columns from your model. 3) Use aggregation tables for large datasets. 4) Avoid complex DAX in calculated columns (use measures instead). 5) Reduce number of visuals per page. 6) Use Performance Analyser (View menu) to find slow visuals.

Power BI Quick Reference

DAX Function	What it does	Simple Example
SUM()	Add column values	SUM(Sales[Revenue])
AVERAGE()	Calculate mean	AVERAGE(Orders[Amount])
COUNT()	Count non-empty	COUNT(Customers[ID])
COUNTROWS()	Count table rows	COUNTROWS(Orders)
DISTINCTCOUNT()	Count unique values	DISTINCTCOUNT(Orders[CustID])
CALCULATE()	Calc with changed filter	CALCULATE(SUM(Sales[Rev]), Year=2023)
FILTER()	Create filtered table	FILTER(Sales, Sales[Amt]>1000)
SUMX()	Row-by-row then sum	SUMX(Sales, Sales[Qty]*Sales[Price])
IF()	Conditional	IF(Score>=90,'A','B')
DIVIDE()	Safe division	DIVIDE(Profit, Revenue, 0)
ALL()	Remove filters	CALCULATE(SUM(...), ALL(Sales))
RELATED()	Get from other table	RELATED(Product[Category])
TOTALYTD()	Year to date	TOTALYTD(SUM(Rev), Dates[Date])
TODAY()	Today's date	DATEDIFF(StartDate,TODAY(),DAY)

CALCULATE is the king of DAX — understand it deeply and you'll crack any Power BI interview! ■